

Does Image Restoration Help Vision-Language Transcription of Historical Modi Manuscripts? A Small-Scale Empirical Study with LipiLens

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Abstract—Modi script manuscripts are largely inaccessible to modern readers and lack mature OCR. We present LipiLens, an AI-assisted pipeline (OpenCV restoration + Qwen2.5-VL-3B with a Modi-specific LoRA adapter + mandatory human verification), and use it to test whether classical image restoration helps vision-language transcription. On 50 real manuscript pages with expert Devanagari ground truth, the baseline character error rate (CER) is 0.304 (range 0.026–0.912). Across a 140-call sweep over seven preprocessing conditions, grayscale conversion is a perfect no-op, denoising, contrast enhancement and deskewing are neutral within noise, binarization clearly degrades accuracy (17 of 20 pages worse, +0.038 mean CER), and the full restoration pipeline performs worst overall (0.359). On clean historical pages, the safest input to the model is the original image.

Index Terms—Modi script, document image restoration, vision-language models, LoRA, optical character recognition, digital preservation.

I. INTRODUCTION

Modi was the administrative script of Maharashtra from roughly the 13th century until it was phased out in favor of Devanagari in the mid-20th century. An estimated tens of millions of documents—land records, correspondence, court papers—survive only in Modi, while the number of fluent readers shrinks. Physical manuscripts are degraded by faded ink, staining, skew and uneven photography, and no mature commodity OCR exists for the script.

General-purpose vision-language models (VLMs), specialized with low-rank adapters (LoRA), now offer draft transliteration of such low-resource scripts, but always require expert review: a fluent, confident model output can be entirely wrong. A second open question is whether classical restoration (denoising, contrast enhancement, binarization, deskewing) helps or hurts a VLM trained on natural-looking images. More preprocessing is not obviously better.

This paper contributes: (1) **LipiLens**, a working end-to-end pipeline—upload, restore, transcribe, verify, archive, search—with structural separation of AI draft and human-verified text; and (2) a small, fully reproducible empirical study (50 pages, 140 inference calls) of restoration effects on transcription accuracy, reported with failures and limitations intact.

II. BACKGROUND AND RELATED WORK

The MoDeTrans dataset [1] provides 2,043 Modi document images with expert Devanagari transliterations across the Shivakalin, Peshwekalin and Anglakalin eras, plus a synthetic companion set (SynthMoDe); the authors also propose MoScNet, a knowledge-distillation VLM framework. Our transcription adapter [2] is a QLoRA fine-tune (rank 32, 1.94% trainable parameters) of

Qwen2.5-VL-3B-Instruct [4] on 5,721 real plus synthetic images. Its author reports test CER 0.328 against a 0.930 zero-shot baseline on 204 held-out real pages. We use 50 training-split rows of [3] with their shipped transliterations as ground truth.

Classical OCR practice applies denoising, contrast enhancement and binarization before recognition; whether these steps transfer to VLM transliteration of historical scripts has, to our knowledge, not been measured for this model and script combination.

III. THE LIPILENS SYSTEM

A React frontend (Transcribe/Library/About) talks to a FastAPI backend over HTTP. Restoration is a modular OpenCV pipeline driven by named configs (Table I); inference sits behind one *transcribe(image, prompt)* interface served locally or from a Colab GPU endpoint; persistence is SQLite (manuscripts, transcriptions, preprocessing configs). The schema makes AI-authoritative use structurally impossible: *ai_transcription* is immutable once written and verification writes only *verified_transcription*.

Config	Stages
original	none (control)
grayscale	grayscale
denoised	+ non-local-means (h=10)
enhanced	+ CLAHE (2.0, 8×8)
binarized	+ Otsu threshold
deskewed	grayscale + Hough deskew
full restoration	all + border cleanup

TABLE I. Preprocessing conditions (cumulative).

IV. EXPERIMENTAL SETUP

Data: 50 MoDeTrans training-split rows (filenames 1.jpg–1042.jpg range), reference lengths 60–205 characters, MIT-licensed; one additional unprovenanced character chart

used only for smoke-testing. Model: Qwen2.5-VL-3B-Instruct at pinned revision 6628554 plus adapter [2] at 5b9957d, 4-bit NF4, served from Colab GPU (type unrecorded). Prompt (from the adapter's official recipe): “Transliterate the text in this image into Devanagari script. Output only the Devanagari text, with no explanation.” Generation: max 256 new tokens, greedy. Metrics: CER and WER via Levenshtein distance against the shipped references. Controls: identical model, prompt and decoding across conditions; only the input variant changes.

Threats: the 50 pages come from the adapter's own training split, so scores may be optimistic; n=50 supports description, not significance testing; pages are relatively clean, short, single-hand excerpts.

V. RESULTS

Baseline (original images, n=50): mean CER 0.304 (std 0.15, range 0.026–0.912), mean WER 0.645. Best page MT-048 reaches 0.026; worst MT-038 reaches 0.912. Substitutions dominate every error profile. Table II and Fig. 1 give the sweep outcome (140 calls).

Condition	n	CER	WER	W/T/L
original	20	0.317	0.688	—
grayscale	20	0.317	0.688	0/20/0

Condition	n	CER	WER	W/T/L
denoised	20	0.321	0.692	8/1/11
enhanced	20	0.328	0.686	5/5/10
binarized	20	0.355	0.723	3/0/17
deskewed	20	0.332	0.699	3/9/8
full restoration	20	0.359	0.728	5/1/14

TABLE II. Sweep results (descriptive; W/T/L = pages better / tied / worse than original at ± 0.005 CER).

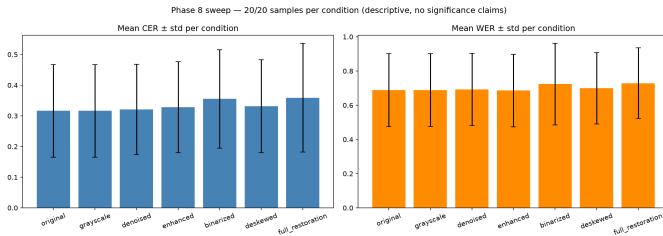


Fig. 1. Mean CER/WER per condition ($\pm$ std). Binarization and the full pipeline sit clearly above baseline.

Three findings stand out. First, grayscale conversion ties the original on all 20 pages to four decimals—a built-in control proving measurement stability. Second, Otsu binarization degrades 17 of 20 pages (+0.038 mean CER); Fig. 2 shows why: thresholding visibly erodes thin strokes, headline rules and joints that the model evidently uses. Third, stacking neutral stages compounds into harm: the full pipeline is worst overall (0.359, 14/20 losses), damaging the cleanest page most (MT-002: 0.099 $\rightarrow$ 0.242). Denoising, enhancement and deskewing sit within noise of baseline, with the largest gains clustering on the highest-baseline pages (e.g., denoised MT-011: 0.550 $\rightarrow$ 0.420). A fluent-but-wrong hallucination on MT-003 (CER 0.567) further justifies mandatory verification.

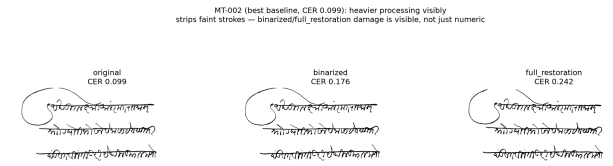


Fig. 2. MT-002 under original / binarized / full restoration. Stroke erosion is visible, not just numeric.

VI. DISCUSSION

H1 (restoration helps) is rejected on these pages; H3 (aggressive processing can hurt, especially binarization) is supported; H2 (effects grow with degradation) gets weak directional support only, since degraded pairs were not tested. The product consequence is immediate: LipiLens defaults to the original image and offers restoration as an option, never the default.

VII. LIMITATIONS

Training-split overlap (optimistic bias); n=50, descriptive only; clean short pages—nothing here covers heavily degraded manuscripts; single prompt and decoding setting; Colab GPU type and VRAM unrecorded; local 4-GB-VRAM inference undemonstrated; ground truth taken on trust from [3].

VIII. CONCLUSION

LipiLens works end-to-end and its embedded study answers the research question for clean Modi pages: send the original to the model, keep the human in the loop, and distrust preprocessing that destroys gray-level stroke detail. All data, code and logs are available for reproduction.

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References.

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